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LIS 510
Mythbusting Assignment #1

<u>“Computers are magical boxes that can do whatever I want it to.”</u>

This myth comes from Bruce Schneier’s “Secrets and Lies,” where he states “People don’t understand computers. Computers are magical boxes that do things. People believe what computers tell them.” Users routinely trust pop-up, login, and system pop-ups without question. People treat computers as authoritative and trustworthy just because they are non-living things that they do not fully understand.

People believe this myth because many tools in the real world are simple and binary. If you flip a light switch, it turns on. If you press on the gas pedal, the car accelerates. People assume computers work the same way, an input is given and an output is received. Cybersecurity attacks come in many forms, but they all have one common denominator. They exploit the blind trust by posing as a friend or an unsuspecting pop-up screen, such as an official looking website, a threat posing as the computer's security, or a phishing email from a firm that the user trusts. All exploits use this myth to take advantage of a users trust to gain their personal information, access to their computer, access to the finances, etc.

Imagine an employee named Nathan. He receives a pop-up on his work laptop stating “Your security certificate has expired. Click UPDATE to restore.” This pop-up looks official and seems like something that the computer “knows.” Nathan clicks. The pop-up wasn’t from his computer, it was malware sitting dormant on his laptop from when he visited a malicious website last week. By trusting this message, he unknowingly installed a remote access tool that gives an attacker full control of his laptop. Nothing about this interaction was suspicious, and as far as Nathan knows, he strengthened the security of his computer while actually handing it over to an attacker. He trusted the “magical box” because he thinks it knows what it’s doing.

This myth can create serious damage to individuals, firms, and large groups of unsuspecting people. People become overtrusting of computer interfaces using names or logos that they recognize. People click buttons and download links because their computer “told them to do it.” There are blind spots in risk perception, as Schneier mentions, people understand obvious physical risks but not subtle digital ones. “People lock their doors and latch their windows…They don’t think that a package could be a bomb, or that the nice convenience store clerk might be selling credit card numbers to the mob on the side.” This myth also creates a vulnerability to social engineering where attackers don’t need to break an encryption or bypass a firewall, they simply ask for your password with a prompt using a name or logo that you trust.

Computers are not magical, they are complex machines following the instructions written by imperfect humans, interacting with other imperfect humans, across networks built by more imperfect humans. A computer does exactly what it is told, but even when the instructions are wrong or malicious. For computers, people are the weakest link. Security needs to adapt to account for human error, not ignore it. A good security outlook requires skepticism, to verify prompts, check URLs, and to confirm with a specialist before installing anything. Computers are powerful, but only as trustworthy as the people who build, configure, and use them.